COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
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Members:	Condesa, Andrie Garcia, Dennis James T. Hare, Andrei Angelo D. Ines, D Jay Larracas, Jerry Matthew Pellazar, Stephen Kurt R.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ASSIGNMENT 9.1 Binary Search Trees

1. PROCEDURE OUTPUT

Objectives: To create a program in C++ using search trees.

1. To be familiarized with the search tree algorithm.

2. To be able to create a program with a search tree algorithm.

Answer the following questions:

1. What is a binary search tree?

A Binary Search Tree (BST) is a type of binary tree where the values are arranged in a specific order. The smaller values are placed on the left side, while the larger values are placed on the right side, which makes it easier and faster to search for a specific value.

2. Where can binary search trees be used?

Binary Search Trees are used wherever data needs to be stored in sorted order while allowing fast lookup, insertion, and deletion.

3. What is a tree traversal?

A tree traversal is a way of visiting every node in a tree exactly once. A tree traversal can follow either an approach such as pre-order, in-order or post-order which dives deep down one branch before backtracking.

4. Differentiate a post-order and pre-order traversal, give examples.

The post-order traversal visits the root node first, then visits the left subtree, and finally the right subtree. On the other hand, the post-order traversal visits the left subtree first, then the right subtree, and the root node.

5. What is a parse tree and its use?

A parse tree or concrete syntax tree is an organized rooted tree data structure that graphically symbolizes a tree-like hierarchical representation of the derivation of a string according to a formal grammar. Visually, a parse tree has many parts. To parse is to break down components (for example, parts of a sentence), and understand their syntactic function and order.

Parse trees is one of the fundamental basis for explaining syntactic structure to source code, assistance in error and syntax detection, some further stages of compilation and lastly applicable in Natural Language Processing, that extracts natural language sentences into hierarchical parts of speech and phrases to help machines comprehend meaning.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

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